SMART FACIAL SCREENING FOR DS DETECTION

Automated AI-assisted screening - not a diagnosis

Patient Information
Name: mart
Date: 2026-03-03 07:58:27
Age: 13
Relation: distant_relative

Screening Summary
Probability: 85%
Confidence: 85%
Risk Level: High
High Confidence: No

Facial Features (AI analysis)

Facial Symmetry: 0.86 Eye Spacing: 0.61 Nasal Bridge: 0.70 Ear Position: 0.56

AI Explanation - ```json | Prediction: 85% | Risk: High

Clinical Impression: This screening indicates High risk with prediction probability 85.0%. Primary Grad-CAM weighted regions are: eyes (37.7%), nose (36.4%), mouth (26.0%). This is a screening interpretation, not a confirmatory diagnosis.

Affected Areas (Point-wise Detailed Analysis):

1. Eyes:
   - Observed (Grad-CAM): intensity 0.628 (high); contribution 37.7%.
   - Why this area matters: peri-orbital spacing/alignment signal weighting.
   - Clinical meaning: is commonly a high-impact region and can significantly increase model confidence when intensity is high.
   - Practical follow-up: prioritize developmental/vision follow-up if broader clinical concerns are present.

2. Nose:
   - Observed (Grad-CAM): intensity 0.606 (high); contribution 36.4%.
   - Why this area matters: nasal bridge and central mid-face structure weighting.
   - Clinical meaning: supports central-face structural pattern detection and often works jointly with eye-region evidence.
   - Practical follow-up: correlate with other central-face signals and pediatric assessment findings.

3. Mouth:

- Observed (Grad-CAM): intensity 0.433 (moderate); contribution 26.0%.
- Why this area matters: lower mid-face contour and oral-facial pattern weighting.
- Clinical meaning: adds lower-midface evidence and is important when combined with nose/chin patterns.
- Practical follow-up: monitor speech-oromotor progress and feeding-related concerns over time.

4. General face:
- Observed (Grad-CAM): intensity 0.0 (low); contribution 0.0%.
- Why this area matters: global face-shape weighting across multiple regions.
- Clinical meaning: indicates diffuse model attention rather than one isolated anatomical hotspot.
- Practical follow-up: reassess with longitudinal monitoring and clinician examination.

Risk Factor Highlights (Percent Contribution):
1. Eyes: 37.7% contribution (intensity 0.628, high).
2. Nose: 36.4% contribution (intensity 0.606, high).
3. Mouth: 26.0% contribution (intensity 0.433, moderate).

Patient Context Influence (derived from background, raw values hidden):
- maternal-age related chromosomal-risk background is elevated; paternal-age related background contribution is mildly elevated; family-relation context suggests higher need for genetic-risk surveillance; first-pregnancy context is considered with neutral baseline weight; adolescent-stage monitoring is emphasized for learning, behavior, and emotional health; reported parental health background is stable; known inherited-disorder background signal appears low; follow-up planning can leverage urban specialist-access pathways.

Patient Background Correlation: The above Grad-CAM region pattern is interpreted together with the provided patient background (child age, parental ages, relation context, and living environment) to prioritize follow-up risk focus.

Future Health Risk Factors (Risk level: High | Probability: 85.0%):
1. Speech and language delay risk may be higher; early speech therapy is recommended.
2. Low muscle tone and motor delay risk may affect posture, gait, and endurance.
3. Feeding/swallow coordination challenges may appear; monitor nutrition and growth closely.
4. Learning and attention difficulties may emerge during school-age years.
5. Mental-health risks include anxiety, low self-confidence, emotional dysregulation, and caregiver stress.

Supporting model narrative: "Pediatric Grad-CAM Screening Explainer", 0.85, "high", ["eyes", "nose", "mouth"], "explanation": "The screening result indicates a high risk of the patient being affected by a genetic condition that can be detected through Grad-CAM analysis.", "risk_factor_explanation": "This is due to the presence of multiple affected regions in the patient's face, which suggests a higher likelihood of inheriting the genetic disorder.", ["food": "broccoli", "reason": "Broccoli is known for its high content of lutein and zeaxanthin, antioxidants that may help protect against some eye diseases." , "food": "carrots", "reason": "Carrots are rich in beta-carotene, which can be converted into vitamin A, a vital nutrient for vision and overall health."], ["exercise": "walking", "reason": "Regular physical activity is important for maintaining good eye health. Walking can help improve. Affected Area Impact (detailed): 1. Eyes: affected intensity 0.628 (high); contribution 37.7%. Eye-region activity often reflects peri-orbital spacing/alignment patterns used by the model. 2. Nose: affected intensity 0.606 (high); contribution 36.4%. Nose-region activity.

Recommended Clinical Follow-up: Pediatric developmental review, speech-language tracking, motor-tone monitoring, sleep/behavior assessment, and caregiver support planning as clinically indicated.

Risk Factor Highlights (Grad-CAM Affected Regions)

- eyes: **27.4%** contribution (high)
- nose: **26.4%** contribution (high)
- mouth: **18.9%** contribution (moderate)
- forehead: **14.7%** contribution (low)
- Patient-context modifier: maternal-age related chromosomal-risk background is elevated; paternal-age related background contribution is mildly elevated; family-relation context suggests higher need for genetic-risk surveillance; first-pregnancy context is considered with neutral baseline weight; adolescent-stage monitoring is emphasized for learning, behavior, and emotional health; reported parental health background is stable; known inherited-disorder background signal appears low; follow-up planning can leverage urban specialist-access pathways.

Future Health Risk Highlights

- Speech and language delay risk may be higher; early speech therapy is recommended.
- Low muscle tone and motor delay risk may affect posture, gait, and endurance.
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- Learning and attention difficulties may emerge during school-age years.
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Recommendations

Food Recommendations

Daily: balanced meals with vegetables, fruit, whole grains, and protein., Weekly plan: include fish/lean protein 2-3 times and reduce sugary snacks.

Exercise Recommendations

Daily: 20-30 minutes of light supervised physical activity., Weekly plan: 5 active days + 2 recovery/stretching days.

Suggested Exercise Videos

- Gentle Child Mobility Routine: https://www.youtube.com/results?search_query=Gentle%20Child%20Mobility%20Routine
- Beginner Facial Muscle Exercise Session: https://www.youtube.com/results?search_query=Beginner%20Facial%20Muscle%20Exercise%20Session

This report is for screening purposes only and is not a diagnosis.